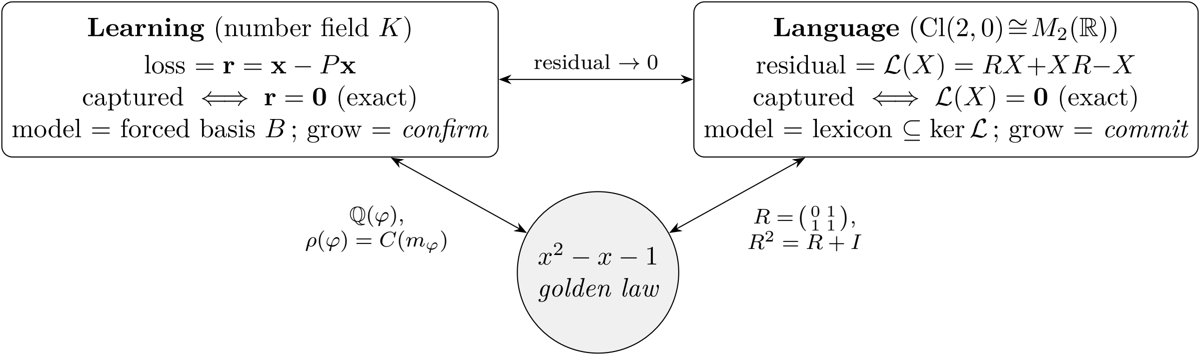
Learning (number field K)
loss $= \mathbf{r} = \mathbf{x} - P\mathbf{x}$
captured $\iff \mathbf{r} = \mathbf{0}$ (exact)
model $=$ forced basis B ; grow $=$ *confirm*
residual $\to 0$
Language ($\mathrm{Cl}(2,0) \cong M_2(\mathbb{R})$)
residual $= \mathcal{L}(X) = RX + XR - X$
captured $\iff \mathcal{L}(X) = \mathbf{0}$ (exact)
model $=$ lexicon $\subseteq \ker \mathcal{L}$; grow $=$ *commit*
$\mathbb{Q}(\varphi)$,
$\rho(\varphi) = C(m_\varphi)$
$R = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$,
$R^2 = R + I$
$x^2 - x - 1$
golden law